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An anti-inflammatory chondroitin sulfate-poly (lactic-co-glycolic acid) composite electrospinning membrane for postoperative abdominal adhesion prevention†

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Postoperative abdominal adhesion is a very common and serious complication, resulting in pain, intestinal obstruction and heavy economic burden. Post-injury inflammation that could activate the coagulation cascade and deposition of fibrin is a major cause of adhesion. Many physical barrier membranes are used to prevent abdominal adhesion, but their efficiency is limited due to the lack of anti-inflammatory activity. Here, an electrospinning membrane composed of poly(lactic-co-glycolic acid) (PLGA) providing support and mechanical strength and chondroitin sulfate (CS) conferring anti-inflammation activity is fabricated for preventing abdominal adhesion after injury. The PLGA/CS membrane shows a highly dense fiber network structure with improved hydrophilicity and good cytocompatibility. Importantly, the PLGA/CS membrane with a mass ratio of CS at 20% provides superior anti-adhesion efficiency over a native PLGA membrane and commercial poly(D, L-lactide) (PDLLA) film in abdominal adhesion trauma rat models. The mechanism is that the PLGA/CS membrane could alleviate the local inflammatory response as indicated by the promoted percentage of anti-inflammatory M2-type macrophages and decreased expression of pro-inflammatory factors, such as IL-1 β , TNF- α and IL-6, resulting in the suppression of the coagulation system and the activation of the fibrinolytic system. Furthermore, the deposition of fibrin at the abdominal wall was inhibited, and the damaged abdominal tissue was repaired with the treatment of the PLGA/CS membrane. Collectively, the PLGA/CS electrospinning membrane is a promising drug-/cytokine-free anti-inflammatory barrier for post-surgery abdominal adhesion prevention and a bioactive composite for tissue regeneration.

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Introduction

Postoperative peritoneal adhesion is an abnormal pathological connection between the peritoneum and surrounding organs or tissues, leading to various complications.¹ Abdominal adhesion is a major complication after abdominal surgery or injury, which causes intestinal obstruction, pain, female infertility and other adverse problems, with incidence of up to 90%.^{2–4} To date, the exact pathophysiology for adhesion occurrence is not fully understood.⁵ Diverse situations, including trauma and excessive inflammation, are involved in the formation of adhesion.⁶ Thereinto, local inflammation has been considered to play an important role in the formation of peritoneal adhesion.^{7,8} Injury caused by surgery or trauma to the abdomen triggers the release of histamine, vasoactive peptides, and various cytokines from stromal cells. Resident macrophages initially adhere and aggregate to damaged tissues, serving as precursors for adhesion development.⁹

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Vasodilators can also be released by recruited macrophages to increase the permeability of blood vessels, resulting in severe fibrous tissue exudation. Ultimately, the balance between the activation of the coagulation system and the fibrinolytic system is destroyed, leading to the production of massive fiber connections at the inflammatory site and the formation of abdominal adhesion.¹⁰

Various clinical therapies, including adhesionlysis surgery, physical barriers, and drug treatment, have been used for abdominal adhesion prevention.^{11–14} However, surgical operation may cause organ damage and bleeding, leading to re-adhesion,¹⁵ and drug treatment also has a limited therapeutic effect.¹⁶ Implantation of commercially available physical barriers, such as hyaluronic acid solution (Interceed®) and sodium hyaluronate carboxymethylcellulose (Seprafilm®), is still the most common method to prevent adhesion, which mainly inhibits the formation of adhesion by simply separating the abdominal wall and the intestine, but exhibits few effects on relieving inflammatory response. Delivery of anti-inflammatory drugs or factors by physical barriers is limited by sophisticated manufacturing, uncontrolled drug release, and potential adverse effects.¹⁷ Therefore, the combination of anti-inflammation and physical separation may represent a promising approach for anti-adhesion, while the efficiency needs to be further improved, without the use of anti-inflammatory drugs or cytokines.

Chondroitin sulphate (CS), a polysaccharide found in cartilage and other tissues in the body,¹⁸ belonging to the family of sulfated glycosaminoglycans (GAGs)¹⁹ with anti-inflammatory, antioxidant and immune enhancing activities,^{20,21} was adopted to rebuild the balance between M1 and M2 phenotype macrophages and modulate the inflammatory tissue micro-environment. Due to its biological activity, CS is widely used to treat osteoarthritis, cardiovascular diseases and tissue adhesion.^{22–24} It was hypothesized that CS could affect the progression of inflammatory response after trauma, which would inhibit the occurrence of tissue adhesion. Hydrogels composed of gellan gum (GLG) and CS were developed as anti-adhesion barriers after abdominal surgery.²⁵ Furthermore, a single intraperitoneal administration of chondroitin sulfate solution could prevent postoperative adhesion formation, by reducing the amount of fibrin and collagen type I deposition.²⁶ Moreover, CS plays important structural and signaling roles, acting as a scaffold and participating in growth factor presentation and binding in peritoneal adhesion prevention.²⁷ In addition to the form of hydrogels and solutions, Seikagaku Corporation (Tokyo, Japan) announced that favorable results were obtained in a pivotal study in Japan for SI-449, a powder adhesion barrier with cross-linked chondroitin as the main ingredient, in the field of gastroenterological surgery in July 2023.²⁴ However, there are few reports on the use of electrospinning technology to combine polymers and CS as a bioactive barrier to prevent postoperative abdominal adhesion.

In this work, we designed an alternative anti-adhesive PLGA electrospun film blended with different contents of CS. The

PLGA membrane acted as the matrix to provide mechanical support and flexibility. PLGA and CS polymers with the mass ratio of 90 : 10 or 80 : 20 were mixed in the solvent and then manufactured into highly fibrous films by electrostatic spinning (Fig. 1). With the increase of CS, the hydrophilicity of the PLGA/CS membrane was improved, but the mechanical properties including Young's modulus and tensile stress were decreased. The obtained PLGA/CS8020 membrane showed superior anti-inflammatory bioactivity in a rat abdominal wall defect model, which promoted the proportion of M2-type macrophages in damaged abdominal tissue and decreased the expression of pro-inflammatory cytokines including TNF- α , IL-1 β and IL-6. It was found that the PLGA/CS membrane suppressed the development of the coagulation cascade and up-regulated the release of tissue plasminogen activator (t-PA). Thus, the fibrinolytic system was activated, and peritoneal adhesion was abolished (Fig. 1). Altogether, anti-inflammation and anti-adhesion PLGA/CS hybrid membranes hold great promise in preventing post-injury tissue adhesion and clinical transformation.

Results

Preparation and characterization of the PLGA/CS membrane

To prepare the PLGA/CS membrane, PLGA and CS were dissolved in HFIP solution containing 10% deionized water,²⁸ at a mass ratio of 90 : 10 or 80 : 20. Electrospinning was used to manufacture the PLGA/CS membrane. The thickness of the membrane was set at 100 μm , which was used in previous anti-adhesion studies.²⁹ When the mass ratio of CS was higher than 20%, the film could not be formed at our presupposed experimental conditions. A PLGA film was constructed as the control. The morphology of the PLGA/CS membrane was characterized by SEM. Fiber diameter and pore size distribution were analyzed using ImageJ software. As shown in Fig. 2a, the structures of PLGA and PLGA/CS membranes were ECM mimics with highly crosslinked fiber networks, and with the increase of CS, the fiber density was increased. In contrast, the surface structure of the commercial PDLLA membrane was melted and sealed (Fig. S1†), which was not conducive to the transmission of nutrients and gas exchange. As the CS ratio was 0, 10% and 20%, the fiber diameter was 1.07 ± 0.1 , 0.25 ± 0.01 and 0.17 ± 0.01 μm (Fig. 2b), respectively. The pore size of the membrane decreased from 5.32 ± 3.8 , 1.17 ± 0.11 to 0.57 ± 0.34 μm with the increment of CS content, suggesting that PLGA/CS membranes could effectively prevent cell infiltration.

The anti-adhesion film should have sufficient mechanical strength to support the injured tissue. The mechanical properties of the membranes were tested using a universal tensile machine. Representative stress–strain curves (Fig. 2c and S2†) demonstrated that Young's modulus of the PLGA/CS8020 membrane (54.17 ± 7.71 MPa) was lower than that of PLGA (300.7 ± 32.95 MPa), the PLGA/CS9010 (149.6 ± 5.23 MPa) membrane and the commercial film (827.3 ± 25.27 MPa) (Fig. 2d and Table S1†), demonstrating that the PLGA/

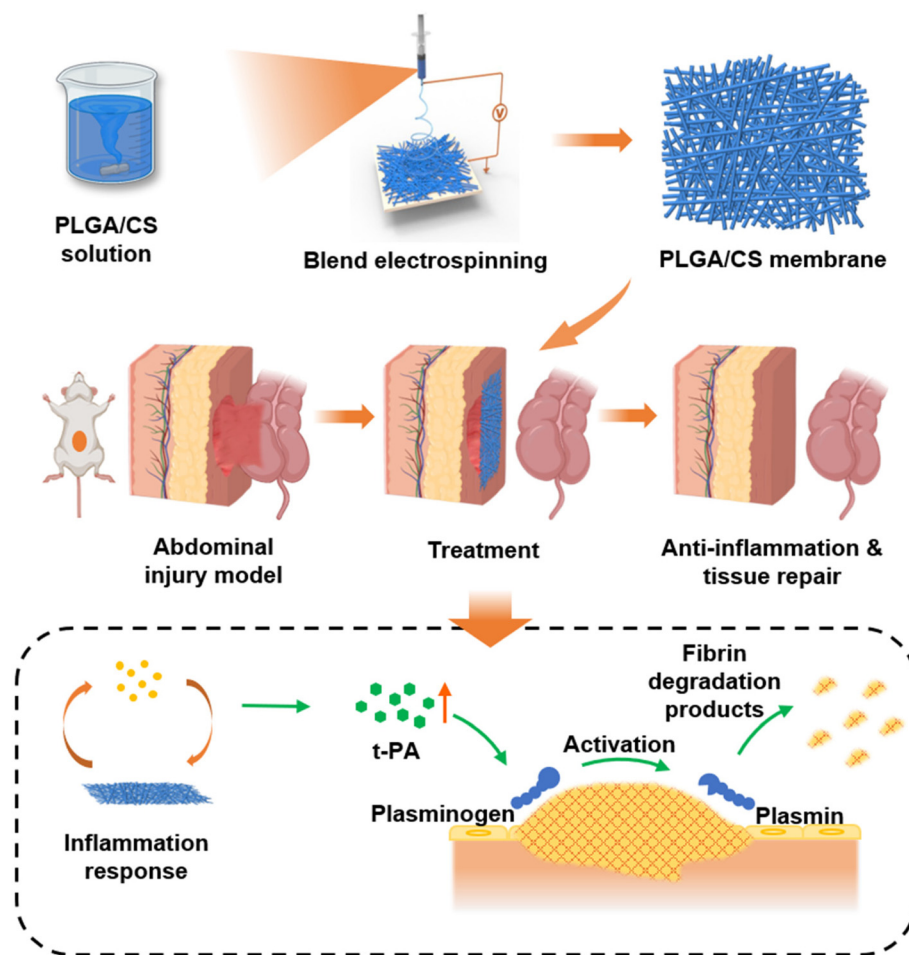


Fig. 1 Schematic illustration of the design and application of the PLGA/CS membrane for preventing abdominal adhesion. The mixed solution of PLGA and chondroitin sulfate was incorporated into the film by electrospinning technology. The PLGA/CS scaffold was implanted at the trauma location to prevent tissue adhesion in abdominal trauma rat models.

CS8020 group had better elasticity than the other groups. The maximum tensile stress and load of PLGA/CS8020 was 1.38 ± 0.03 MPa and 2.04 ± 0.19 N, respectively, which were lower than that of PLGA (6.8 ± 0.93 MPa, 8.17 ± 1.12 N), PLGA/CS9010 (3.76 ± 0.73 MPa, 4.51 ± 0.87 N) and the commercial film (9.64 ± 0.68 MPa, 3.86 ± 0.27 N) (Fig. 2e and f and Table S1†). As the CS ratio increased from 0 to 10% and 20%, the failure strain decreased from $366.3 \pm 21.55\%$ to $164.5 \pm 42.3\%$ and $24.3 \pm 10.76\%$ (Fig. 2g). However, the failure strain of the PLGA/CS8020 membrane was still higher than that of the commercial membranes ($5.91 \pm 1.08\%$) (Table S1†). These results suggest that PLGA/CS films have favorable mechanical properties for anti-adhesion application. Water contact angle is an important index to judge the hydrophilicity of materials. The contact angle of PLGA, PLGA/CS9010, the PLGA/CS8020 film and the commercial film was $123.7 \pm 0.76^\circ$, $116.1 \pm 12.66^\circ$, $74.1 \pm 7.88^\circ$ and $78.3 \pm 9.01^\circ$ (Fig. 2h and Table S1†), respectively. The addition of CS improved the hydrophilicity of polymer membranes, and transformed the hydrophobic nature of the native PLGA membrane into hydrophilic.

***In vitro* cytotoxicity, cell adherence, enzymatic degradation and anti-inflammation**

Live/dead staining was used to evaluate the toxicity of various membrane extracts against L929 fibroblasts. The membranes were immersed in DMEM medium for 48 hours at 37°C to obtain the extract, which was cultured with cells. Live/dead staining showed that no cytotoxicity to L929 cells was observed in each group with the treatment of membrane extracts and no significant difference was indicated within the culturing time period of 3 days (Fig. 3a and S3†). Fibroblasts showed a typical spindle shape with good vitality. Moreover, the cell density of the PLGA/CS9010 and PLGA/CS8020 membrane groups was comparable with that of the control group or the PLGA group, indicating that the addition of CS did not affect cell growth. To further explore cell proliferation, a CCK-8 assay was performed to detect the cell viability. The absorbance was detected using a microplate reader at 450 nm. Cells in all groups proliferated as time was extended, and there was no significant difference between the control and experimental groups (Fig. 3b and

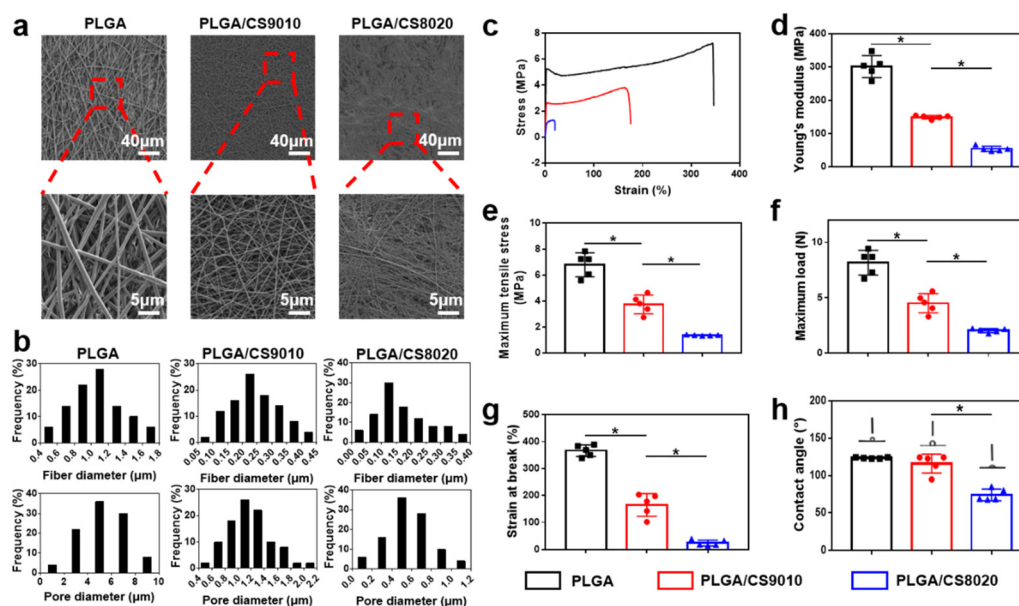


Fig. 2 Characterization of the PLGA/CS membrane. (a) Surface morphology of the PLGA/CS electrospun membrane. (b) Fiber and pore diameter distributions in the PLGA/CS membranes. (c) Stress–strain curve, (d) Young's modulus, (e) maximum tensile stress, (f) maximum load, (g) strain at break, and (h) contact angles of diverse membranes. Data are presented as mean $\pm$ SD ($n = 5$).

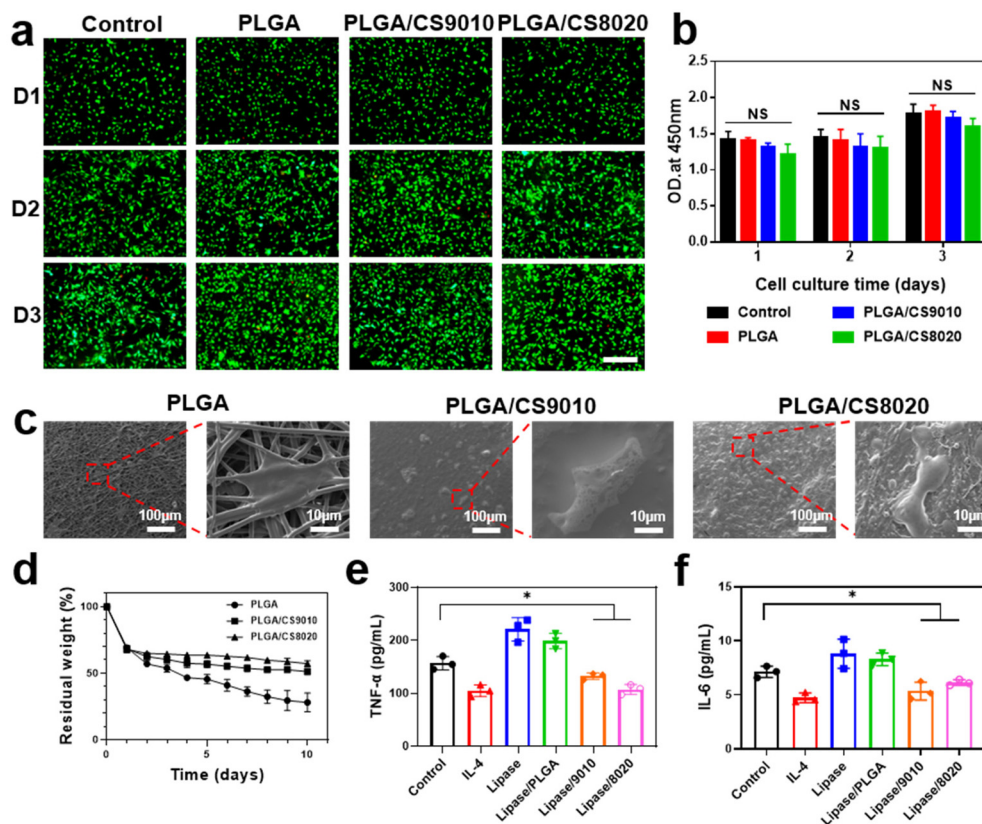


Fig. 3 Cytotoxicity of diverse films and cell adhesion. (a) Live/dead staining and (b) CCK-8 assay. Scale bar, 200 μ m. Data are presented as mean $\pm$ SD ($n = 5$). (c) Bioelectronic microscopy of cells on the membrane. (d) Degradation curve of the PLGA, PLGA/CS9010 and PLGA/CS8020 membranes. (e) TNF- α and (f) IL-6 levels in BMDMs simulated by various membrane degradation products. Data are presented as mean $\pm$ SD ($n = 3$).

S4†). Then, the morphology of cells on the membrane was observed by inoculating HUVECs on the membrane at a density of 50 000 cells per well. Cells were fixed and freeze-dried, and subjected to freeze electron microscopy (Fig. 3c and S5†). The PLGA membrane was hydrophobic, and cells had difficulty to attach and spread on it. However, cells could easily spread on the PLGA/CS9010 and PLGA/CS8020 membranes, due to the hydrophilic modification. The PLGA/CS8020 membrane was almost fully covered by cells, suggesting promoted cell adhesion, which may be conducive to wound healing.³⁰

An enzymatic degradation experiment was carried out in PBS containing type II lipase. As shown in Fig. 3d, all membranes showed burst degradation on the first day, due to the activity of lipase. The mass of the PLGA film decreased to $45.56\% \pm 2.78\%$ and $28.02\% \pm 5.78\%$ of its initial mass after 5 and 10 days, respectively. In contrast, the PLGA/CS membrane degraded slowly after the initial burst degradation. At day 10, a weight loss of $48.79 \pm 1.78\%$ and $42.82 \pm 1.84\%$ was observed for the PLGA/CS9010 and PLGA/CS8020 groups, respectively. The PLGA/CS8020 membrane exhibited a lower weight loss during enzyme degradation at each time point compared to PLGA/CS9010 or PLGA, which was similar to such hybrid membranes added with other GAGs, such as hyaluronic acid.^{31,32} Moreover, the degradation rate of the PLGA/CS membrane decreased with the increase of CS content, indicating that the addition of CS was able to adjust the degradation rate of the hybrid membrane.

The degradation products of PLGA are described as non-toxic. However, the release of lactate will cause a local acidic microenvironment, which can induce inflammation.³³ To verify the anti-inflammatory ability of CS in a PLGA degradation environment, we measured the levels of pro-inflammatory cytokines secreted by bone marrow-derived macrophages

(BMDMs) stimulated by membrane degradation products at day 3 (Fig. 3e and f). Compared to the control group, the stimulation of PLGA degradation products increased the expression of TNF- α and IL-6, while the treatment of PLGA/CS9010 and PLGA/CS8020 membrane products significantly reduced the levels of these pro-inflammatory factors, indicating that CS endowed the membrane with notable anti-inflammatory effects in the context of PLGA degradation. In addition, we also validated the effect of native CS on the production of pro-inflammatory cytokines in BMDMs (Fig. S6†). It was found that an increase in CS concentration reduced the secretion of TNF- α and IL-6, directly indicating the essential anti-inflammatory effect of CS.

In vivo anti-adhesion using a rat abdominal wall defect model

We first evaluated the anti-adhesion efficacy of the PLGA/CS membrane in a rat abdominal wall defect model. The muscular layer of the abdominal wall was punched and a hole was drilled with a diameter of 6 mm. Then the four corners of the anti-adhesion membrane with a size of 2×2 cm were fixed around the wound. On the 7th and 21st day after the treatment, the injured tissues were observed and collected for pathological analysis (Fig. 4a). A standard adhesion scoring system was used to grade the adhesion at the injured tissue: score 0, no adhesion; score 1, mild, easily separable intestinal adhesion; score 2, moderate intestinal adhesion, separable by blunt dissection; and score 3, severe intestinal adhesion requiring sharp dissection to separate.³⁴ Through general observation, the cecum and abdominal wall of the control group were seriously adhered on the 7th day, and the cecum even filled the wound on the abdominal wall after 21 days (Fig. 4b). For the PLGA group, no matter at day 7 or 21, the cecum or small intestine adhered closely to the membrane, and the adhesion score was graded within 2–3. Treatment with

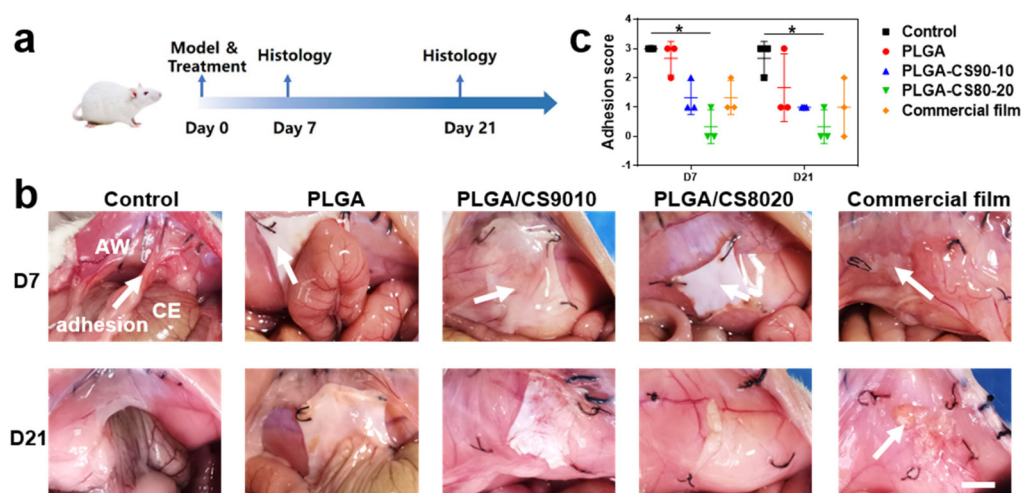


Fig. 4 Anti-adhesion effect against abdominal wall–cecal adhesion model in rats. (a) Schedule for the postoperative anti-adhesion. (b) The implantation of the PLGA membrane, commercial film, and PLGA/CS membrane to prevent the development of tissue adhesion. AW: abdominal wall; CE: cecal mucosa. Scale bar, 1 cm. The white arrow indicates the anti-adhesive membrane. (c) Distribution of adhesion scores in different groups at days 7 and 21 after surgery. Data are presented as mean $\pm$ SD ($n = 3$).

the commercial PDLLA membrane could decrease the abdomen–cecum adhesion, with lower scores than the control group, but adhesion was still found in several animals. Significantly, the PLGA/CS8020 treatment showed a superior anti-adhesion effect over the PLGA membrane and commercial film, and 2 out of 3 rats showed no observable adhesion (score 0), while only 1 rat showed mild and easily separable intestinal adhesion on the 7th day and the 21st day (Fig. 4c).

Hematoxylin and eosin (H&E) staining and Masson's trichrome (MT) staining were performed to further analyze the adhesion formation and collagen deposition (Fig. 5a). In the control and PLGA groups, there were severe adhesion and massive collagen deposition between the abdominal wall and the intestine on the 7th day and the 21st day. Besides, a large number of inflammatory cells infiltrated in the adhesive tissue, suggesting severe inflammatory responses at day 7. In contrast, there were less adhesion in the PLGA/CS9010 and PLGA/CS8020 groups, and only a minimal amount of collagen tissues was observed (Fig. 5b and c). Surprisingly, the muscle

layer in the abdominal wall was almost completely repaired with a clear muscle structure without excessive collagen deposition. However, the damaged tissues in the commercial membrane treatment group were not fully repaired and a large number of collagens was deposited. Furthermore, by comparing PLGA/CS8020 and PLGA/CS9010, it was evident that with the increase of CS content in the composite membrane, wound healing and anti-adhesion efficacy were both improved.

The expression of anti-inflammation cytokines and macrophage polarization

Inflammatory reaction was considered to be the main cause of adhesion.^{35,36} When injured by surgery, the body releases histamine, vasodilators and various inflammatory factors.^{10,37} When the balance of the fibrinolytic system is broken by inflammation and the deposition of fibrin is higher than the degradation, a fibrous band would be formed, causing abdominal adhesion.^{38,39} We further explored the mechanisms of adhesion prevention using a PLGA/CS membrane. The

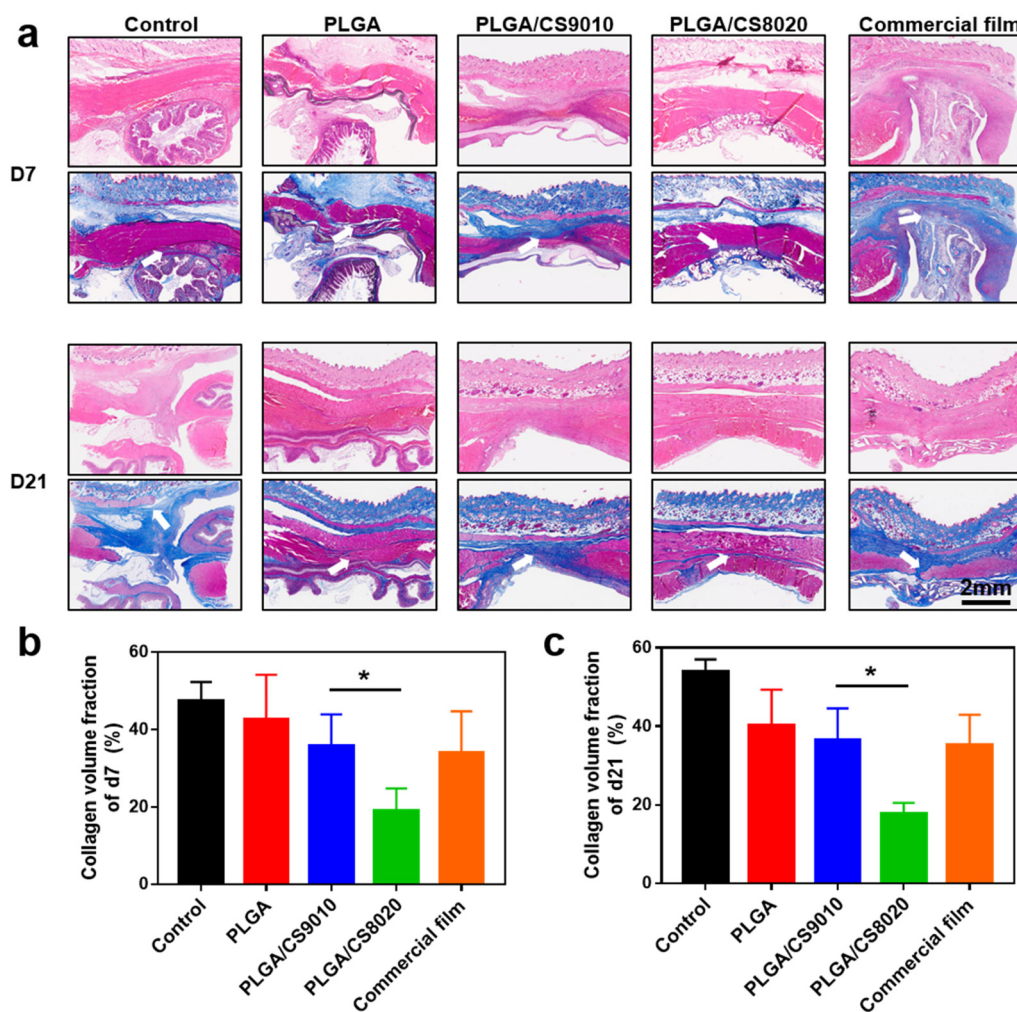


Fig. 5 Representative images for (a) hematoxylin and eosin (H&E) and Masson's trichrome staining (MT). The white arrow represents the damaged area. Volume fraction of collagens in the adhesive tissues from different groups on (b) days 7 and (c) 21 after surgery. Data are presented as mean $\pm$ SD ($n = 3$).

expression levels of pro-inflammatory factors (TNF- α , IL-1 β and IL-6) and anti-inflammation factors (TGF- β , IL-4 and IL-10) were detected using an ELISA kit. Fig. 6a–c show that on the 7th day post-surgery, inflammatory factors were expressed at a high level in the acute inflammatory phase. However, the pro-inflammatory cytokines including IL-1 β , TNF- α , and IL-6 in the PLGA/CS8020 group were significantly lower than those in the control group. For instance, the secretion of TNF- α was significantly decreased from 182.8 ± 19.18 pg mL $^{-1}$ in the control group to 124 ± 24.03 pg mL $^{-1}$ in the PLGA/CS8020 group. For anti-inflammatory cytokines such as IL-4, IL-10 and TGF- β (Fig. 6d–f), implantation of the PLGA/CS8020 membrane increased their expression by 30.3%, 12.3% and 29% compared with those in the control group, respectively, whereas the enhancement percentage in the PLGA/CS9010 group was 13%, 24.7% and 22.9%, respectively. As time was extended to 21 days after operation, typically corresponding to the later

stage of repair, the overall level of inflammatory factors was lower than that on the 7th day. The levels of pro-inflammatory factors in the control group and PLGA group were still high, indicating the presence of chronic inflammation. However, pro-inflammatory cytokines such as IL-1 β , TNF- α and IL-6 in the PLGA/CS8020 group were significantly decreased by 23.4%, 36.4% and 67.6% compared with those in the control group (Fig. 6g–i). The anti-inflammatory factors such as IL-4, IL-10 and TGF- β in the PLGA/CS8020 group were increased by 45.9%, 28.4% and 58.6%, respectively, compared with the control group, while the PLGA/CS9010 group showed a reduction of 27.7%, 36.8% and 49.7% (Fig. 6j–l). In particular, the PLGA/CS8020 membrane resulted in a significantly lower level of IL-6 in comparison with the PLGA/CS9010 membrane. At day 21, treatment with the PLGA/CS8020 membrane notably induced a higher secretion of IL-4, IL-10, and TGF- β , either in comparison with PLGA/CS9010 and PLGA membranes or the

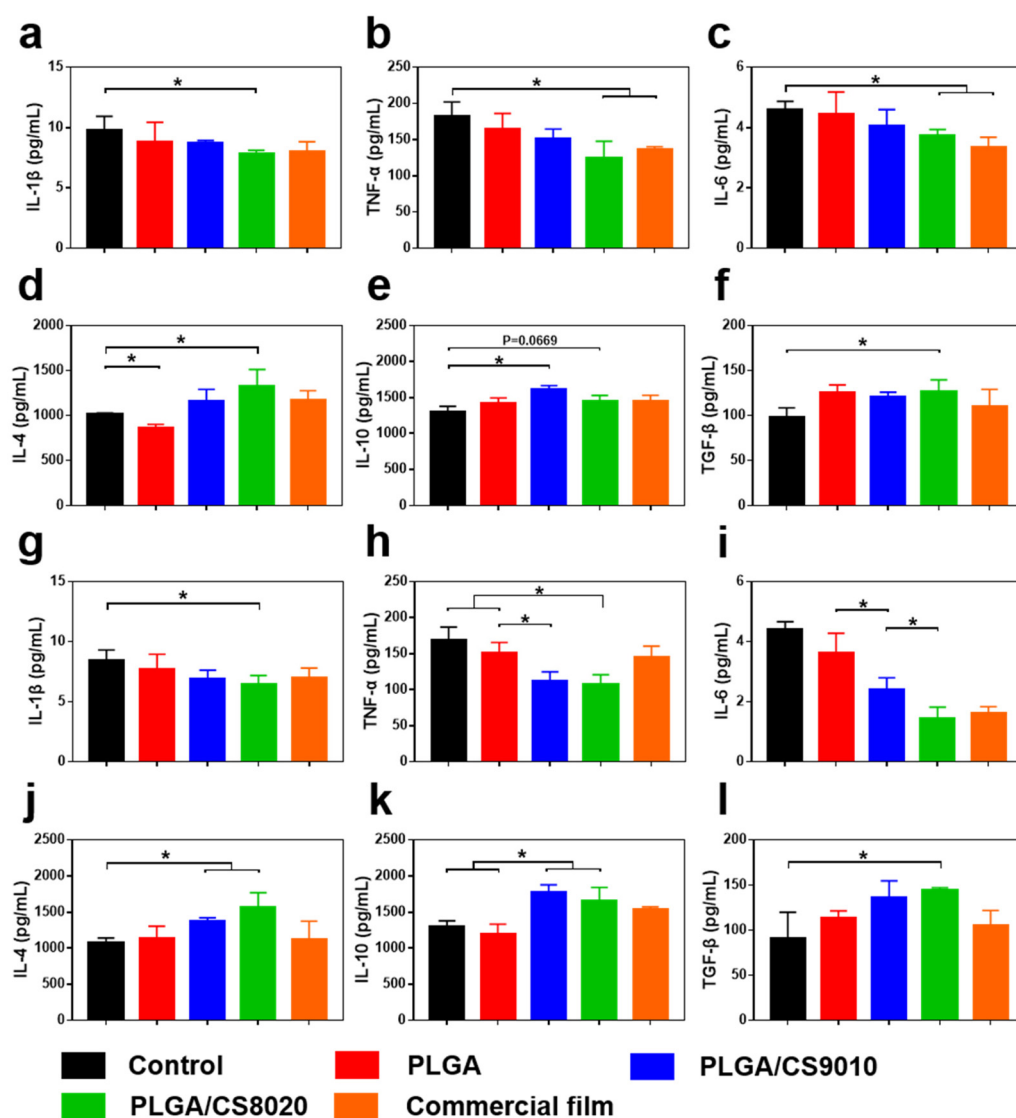


Fig. 6 The release of pro- and anti-inflammatory cytokines at day 7 (a–f) and day 21 (g–l) post injury. Data are presented as mean $\pm$ SD ($n = 3$).

control. These results indicated that the PLGA/CS8020 membrane could promote the expression of anti-inflammatory factors in both the acute inflammatory stage and the tissue repair phase, thereby reducing the inflammatory response.

Macrophage phenotype is also an important indicator for immune response.⁴⁰ M2 macrophages, acting as an anti-inflammatory phenotype, play a vital role in immunoregulation and wound healing.⁴¹ To identify the proportion of M2 macrophages in monocytes, we performed immunofluorescence staining for CD68 and CD206 against tissues collected from adhesion sites on days 7 and 21 after surgery (Fig. 7a and S7[†]). After being dewaxed, rehydrated and antigen repaired,

paraffin sections were immunocytochemically stained for macrophage phenotype markers including CD68 as a pan-macrophage lineage marker and CD206 as an anti-inflammatory phenotype marker (M2) to quantitatively analyze macrophage polarization. In general, the number of CD68⁺ macrophages on the 21st day was lower than that on the 7th day, indicating that the switch of the acute inflammatory stage to the tissue repair stage. As shown in Fig. 7, the PLGA/CS8020 treatment significantly increased the percentage of anti-inflammatory M2 macrophages both on the 7th day and the 21st day, in comparison with all other treatments. These data collectively suggested that the PLGA/CS8020 membrane could contribute to anti-inflammation as indicated by the increase of

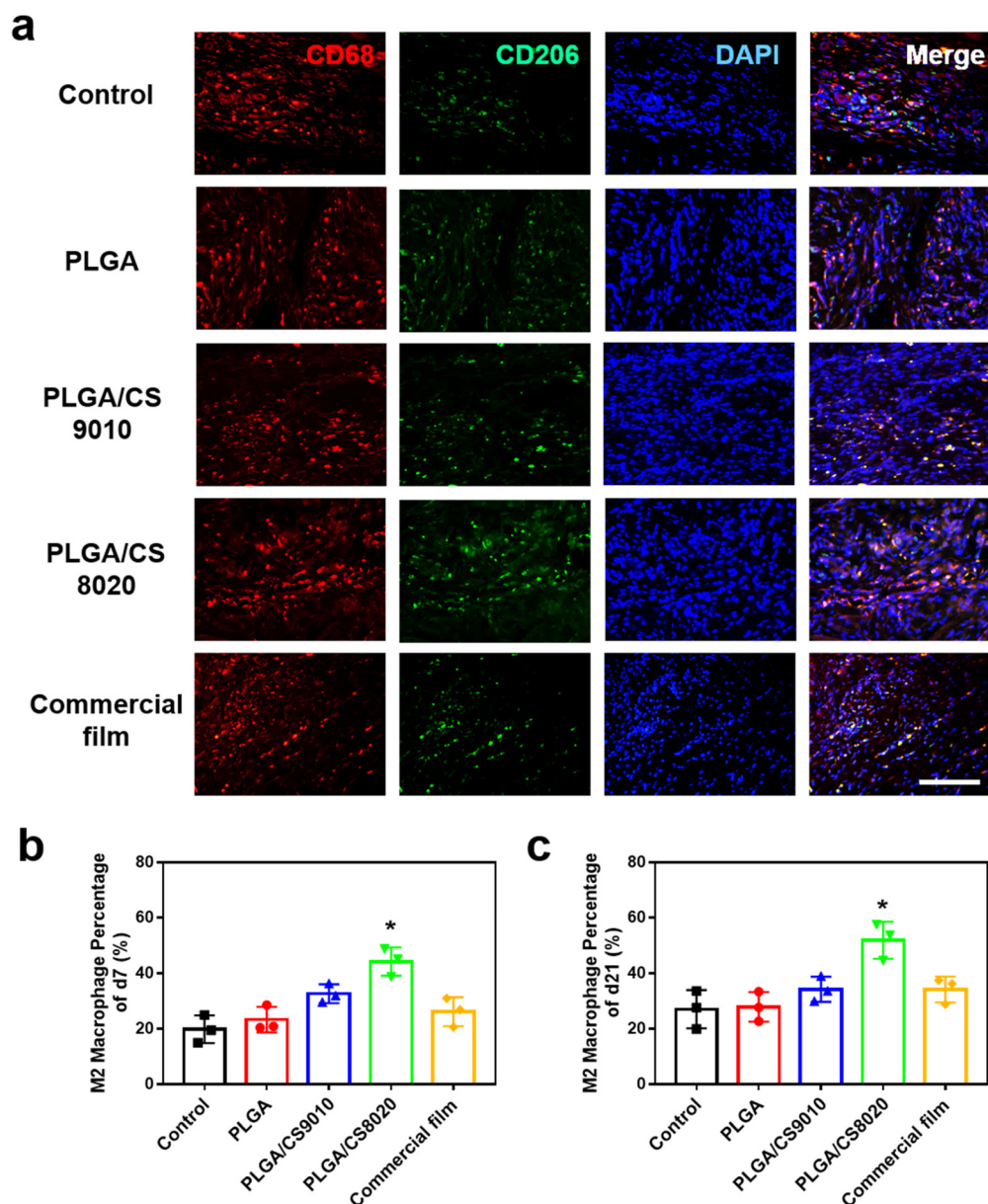


Fig. 7 The PLGA/CS8020 membrane induced the increase of M2-type macrophages. (a) Representative immunofluorescence staining images of CD68/CD206 on day 21 post surgery. Scale bar, 100 μ m. (b and c) Statistical data of the percentage of M2 macrophages, $n = 3$.

M2 phenotype macrophages and the reduction of inflammatory cytokines.

In vivo fibrinolytic activity of PLGA/CS membranes

To further explore the anti-adhesion mechanism, immunofluorescence staining was implemented to detect the activity of the fibrinolytic system (Fig. 8). Tissue plasminogen activator (t-PA) is the key mediator of fibrinolysis, while the plasminogen activator inhibitor (PAI)-1 is the major inhibitor of t-PA, leading to the reduction of fibrinolytic activity.⁴² The balance between the expression of t-PA and PAI-1 affects the activation or inhibition of the fibrinolytic system. We evaluated the protein expression of t-PA and PAI-1 at 21 days post-surgery (Fig. 8a). The expression of t-PA in the PLGA/CS8020 group was obviously higher than that in the control, PLGA and commercial groups, while PAI-1 was inhibited (Fig. 8b and c). The results demonstrated that the fibrinolytic system at the injured tissue was activated by the PLGA/CS8020 anti-adhesion membrane, thus promoting the dissolution of the abnormal adhesive tissue.

Discussion

Abdominal adhesion is a common postoperative complication, which occurs after surgery or abdominal injury.⁴³ The damaged lesion and surrounding tissues or organs are abnormally connected by fiber bridges,¹⁰ caused by excessive inflammation. The formation of adhesion will affect the function of the intestine, leading to intestinal obstruction, infertility, and even a second operation.⁴⁴ With increasing attention being paid to abdominal adhesions, many techniques such as physical isolation and hydrogel and drug therapy are widely used to

prevent adhesion. However, most traditional anti-adhesion membranes suffer from poor hydrophilicity, no anti-inflammatory effect, and even hold the possibility of eliciting inflammatory reaction. Although anti-inflammatory drugs can be loaded into anti-adhesion membranes, the dosage and release of drugs need to be carefully designed, and the side effects of drugs or cytokines should also be seriously considered. These issues could confine the translation of drug- or cytokine-encapsulated anti-adhesion membranes as medical devices. Electrospinning is a common method for manufacturing nanofibers for anti-adhesion, drug release, tissue engineering, and so on.^{45–48} In this work, we designed a PLGA electrospun membrane loaded with chondroitin sulfate (CS) to prevent postoperative abdominal adhesion. The PLGA membrane mainly prevented the connection between damaged tissues and surrounding organs, playing a role of mechanical support. CS, as a multifunctional signaling molecule, played a role in reducing the inflammatory response during the repair process of damaged tissues.⁴⁹ PLGA/CS membranes were prepared by electrospinning technology after simply blending PLGA and CS. Such a hybrid membrane holds merits including low cost, easily available raw materials, as well as simple manufacturing procedures, which are conducive to large-scale production of anti-adhesion film products for clinical application.

Initial acute inflammatory response is a normal physiological phenomenon, but excessive inflammation contributes to the formation of abdominal adhesion.⁵⁰ We evaluated the anti-inflammatory and anti-adhesion effects of our PLGA/CS membrane, in comparison with the commercial PDLLA film. Macrophages may remarkably contribute to the development of adhesion.⁹ Data shown in Fig. 7 clearly suggest that the anti-adhesion effect of PLGA/CS is quite related to the presence of M2-type macrophages in adhesive tissues. A higher

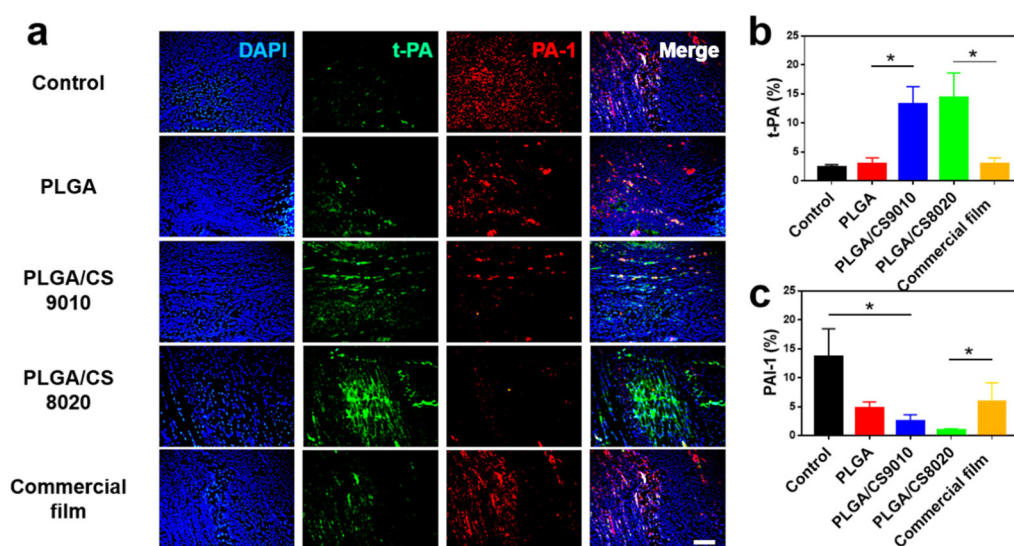


Fig. 8 The activation of the fibrinolytic system by the PLGA/CS membrane. (a) Representative immunofluorescence staining images of different treatments on day 21. (b and c) The concentrations of t-PA and PAI-1 in the tissues surrounding the abdomen and cecum on day 21. Data are presented as mean $\pm$ SD ($n = 3$). Scale bar, 100 μ m.

ratio of anti-inflammatory M2 macrophages could result in a better anti-adhesion effect. *In vitro* anti-inflammation induced by CS alone or PLGA/CS membrane degradation products also demonstrated the anti-inflammatory function of CS. These results together indicate that the anti-adhesion effect of the PLGA/CS membrane may be triggered by macrophage polarization to anti-inflammatory subtype. The accumulation and differentiation of macrophages after the model-establishing surgery, which possibly impacts adhesion formation, need to be further elucidated in future. After abdominal injury, the balance between fibrinolysis and deposition was broken by a coagulation cascade and an inflammatory reaction.⁵¹ Excessive release of pro-inflammatory factors, including IL-1 β , TNF- α and IL-6, upregulated the secretion of PAI-1 and triggered the activation of the coagulation system and inhibition of the fibrinolytic system, resulting in the formation of adhesion.³⁵ The implantation of the PLGA/CS membrane significantly downregulated the expression of PAI-1 and upregulated the expression of t-PA. The fibrinolytic system was activated, which promoted the transformation of plasminogen into plasmin and dissolution of fibrin, as proven by the secretion of matrix metalloproteinase 9 (MMP-9).³⁹ For the structure–property relationship, the PLGA/CS8020 group resulted in the highest level of t-PA, showing the best anti-adhesion effect. Furthermore, we can also conclude that the anti-inflammatory effect of PLGA/CS may help the activation of the fibrinolytic system. Collectively, the PLGA/CS8020 membrane can dramatically decrease the inflammatory response caused by injured peritoneum and inhibit the formation of abdominal adhesion, which allows the anti-adhesion membrane to be applied in various inflammation-triggered adhesion scenarios caused by tissue injury, such as myocardial infarction, tendon and epidual damage, etc.

In addition to their anti-adhesion properties, it was surprisingly discovered that PLGA/CS membranes were able to promote tissue repair. It should be attributed to the fact that anti-inflammatory and pro-regenerative macrophages recruit fibroblasts and reparative cells, followed by the secretion of ECM, eventually leading to tissue remodeling.^{41,52} Furthermore, the hydrophilic PLGA/CS membranes potentiate cell adhesion and proliferation at the impaired tissue site, which is beneficial for wound healing. Moreover, better hydrophilicity enables the membrane to adhere closely to the tissue surface, which is conducive to the release of anti-inflammatory components.⁵³ Several previous literature studies have reported that CS played a direct role in inhibiting adhesion of inflammatory cells to fibronectin matrices and type I collagen;⁵⁴ however, the anti-inflammatory mechanism in anti-adhesion needs to be further validated by tracking the evolution of inflammation during the whole period of anti-adhesion. Furthermore, deep research for CS is needed on the repair of damaged tissues through promoting cell adhesion, migration, and differentiation.

In terms of the limitations of this study, a typical electrospinning approach was performed to fabricate a PLGA/CS membrane, using volatile organic solvents, such as 2,2,2-tri-

fluoroethanol or 1,1,1,3,3,3-hexafluoro-2-propanol to dissolve polymers. Residual organic reagents affect the mechanical properties of the materials and increase potential cytotoxicity, and need to be fully removed, or an alternative “green” solvent should be used as an alternative. In addition, with the increase of CS content, the mechanical strength of the anti-adhesion film decreases. The method to maintain or reduce the loss of the mechanical properties of the hybrid membrane is also an issue that needs to be studied. Additionally, with the popularity of non-invasive surgery, the seamless suture strategy will be prevalent for *in vivo* transplantation. Increasing the self-adhesive feature of the membrane to achieve suture-free surgical operation will be valuable in future research. The preparation of composite membranes combined with a type of adhesive hydrogel could be adopted to further widen the clinical application scenarios, including but not limited to adhesion prevention and tissue regeneration.

Experimental

Materials

The PLGA copolymer (molar ratio of LA to GA = 50 : 50) was purchased from Jinan Daigang Biomaterials Co., Ltd. Chondroitin sulfate sodium salt (CAS: 39455-18-0) was purchased from Sigma-Aldrich (Shanghai, China). Commercial PDLA films (thickness: 0.04 mm) were purchased from Guangzhou Hongjian Biomedical Products Co., Ltd. L929 and HUVEC cells were purchased from the Cell Bank of China Academy of Sciences, and cultured according to the manufacturer's guidelines. Briefly, L929 cells were maintained in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (FBS), 100 U mL⁻¹ penicillin, and 100 U mL⁻¹ streptomycin, which were cultured at 37 °C and under 5% CO₂ in an incubator. HUVEC cells were cultured in endothelial cell medium (SKU: 1001, Sciencell) with 0.05 mg mL⁻¹ Endothelial Cell Growth Supplement (ECGS), 10% FBS and 1% penicillin–streptomycin (P/S) in a standard culture environment of 37 °C and 5% CO₂.

Preparation and characterization of the PLGA/CS membrane

PLGA and CS with a mass ratio of 90/10 or 80/20 were dissolved in 1,1,1,3,3,3-hexafluoroisopropanol (HFIP) containing 10% deionized water and stirred at room temperature for 24 hours. The PLGA/CS membranes were prepared using an electrospinning machine (type: YFSP-T, Yunfan (Tianjin) Instrument Co., Ltd). The solution was loaded into a 10 mL plastic syringe with a blunt stainless-steel needle (20 G) and fed at a speed of 2 mL h⁻¹. The needle was placed 15 cm away from the cylinder receiver, which was covered with a layer of aluminum foil and rotated at 500 rpm. A high voltage of 16 kV was applied between the tip of the needle and the cylinder collector. The thickness of the PLGA/CS membrane was controlled at 100 μ m. After manufacturing, the PLGA/CS membrane was removed from the aluminum foil and placed in the fume hood at room temperature for 48 hours.

Scanning electron microscopy (SEM) was used to investigate the morphology of the PLGA/CS membrane. Before observation, a sputtering coater (SC7620, Quorum, England) was used to spray gold on the sample for 45 s with the gold spraying electricity of 10 mA. Then, the surface of the PLGA films was observed by SEM (MIRA LMS; Tescan, Czech Republic) at an acceleration voltage of 3 kV. Fiber diameter and pore size were quantified using ImageJ software and 50 fibers or pores were randomly selected for statistical analysis. The water contact angles were measured using an optical contact angle meter (OCA20, Data Physics Corporation, Germany). The mechanical strength of the membranes was measured using a universal tensile machine (Instron, USA). Samples were cut into 1×5 cm strips and stretched at a speed of 10 mm min^{-1} until they were destroyed.

Enzymatic degradation experiments of the PLGA/CS membranes were characterized by a complete loss of weight. Each membrane (1×1 cm) was immersed in 20 mL of PBS containing 1 mg mL^{-1} lipase and incubated at 37°C . At regular times, residual weight of each membrane was weighed after being dried.

The *in vitro* anti-inflammatory effect of the PLGA/CS membranes was investigated by measuring the cytokine levels in BMDM using an ELISA kit. BMDMs were extracted from C57BL/6 mice according to a previous study.⁵⁵ Then, BMDMs of each mice were evenly seeded in a 6-well plate with 2 mL of RPMI 1640 medium containing 10% inactivated FBS and 20 ng mL^{-1} M-CSF. After 6 days of cultivation, BMDM was stimulated by the sterilized membrane degradation solution of each group on the 3rd day and CS with gradient concentration. Two days later, an ELISA kit was used to detect the levels of TNF- α and IL-6 of in the supernatant of the culture medium in each group.

In vitro cytocompatibility

Live/dead staining was used to detect the toxicity of the extracts of diverse membranes. Briefly, the PLGA or PLGA/CS membranes were incubated with 5 mL of DMEM medium containing 10% FBS at 37°C . The ratio of medium to membrane was 5 mL cm^{-2} . After incubation for 48 hours, the membrane was removed, and the obtained extract was stored at 4°C for the following study. L929 cells were seeded in a 24-well plate at a density of 5000 cells per well with 1 mL of DMEM medium containing 10% FBS and 1% penicillin–streptomycin per well. After culturing for 24 hours, the medium was replaced with 1 mL of DMEM medium containing 20% extract, and an equal amount of medium without extract was added to the control group. After incubating for 1, 2, and 3 days, the medium was discarded, and 250 μL of a detection working solution containing 1 μL of calcein AM and 1 μL of PI was added into each well. Afterward, cells were incubated for 30 minutes at 37°C in the dark, and observed using an inverted biological microscope (DMI8 manual, Leica, Germany) under excitation light at 494 or 535 nm.

The activity of fibroblasts was further verified using a cell counting kit-8 (CCK-8) assay. Briefly, L929 fibroblasts were

seeded into a 96-well plate at a density of 2000 cells per well and cultured in a standard culture environment of 37°C and 5% CO_2 . After 24 hours, the extract diluted to 20% by DMEM medium was added into each well, and the medium without the extract was taken as the blank control. The medium containing the extract was changed every day. After being cultured for 1, 2 or 3 days, the medium in each well was replaced by 200 μL of DMEM medium containing 10 μL of CCK-8 solution. Then, cells were incubated for 1 hour and measured using a microplate reader (3001, Thermo Fisher Scientific) at a wavelength of 450 nm.

To further investigate the cytocompatibility, human umbilical vein endothelial cells (HUVECs) were seeded on the PLGA/CS membrane. Cells were cultured in the endothelial cell medium with 10% FBS. Membranes were pasted on the bottom of the 12-well cell culture plate and sterilized by ultraviolet light for 24 hours. Then, HUVECs were seeded on the membrane at a cell density of 50 000 cells per well. After culturing for 24 h, cells were fixed with 2.5% glutaraldehyde for 3 hours and washed twice with PBS. Cells were dehydrated using ethanol with a concentration gradient of 30, 50, 70, 80, 90 and 100%, and then freeze-dried. Finally, cells were observed using a SEM at an accelerating voltage of 3 kV.

In vivo anti-adhesion evaluation

Sprague-Dawley (SD) rats (male, 200–250 g) were obtained from Beijing Vital River Laboratory Animal Technology Co., Ltd. All animal procedures were performed in accordance with the Guidelines for the Care and Use of Laboratory Animals of Peking Union Medical College and approved by the Animal Experiments and Ethics Review Committee of the Institute of Radiation Medicine, Chinese Academy of Medical Sciences (approval no: SYXK (Jin) 2019-0002). Thirty SD rats were divided into 5 groups, including control, PLGA, PLGA/CS9010, PLGA/CS8020 and commercial film groups. The rats were anesthetized with isoflurane and then continued to be anesthetized with a rat ventilator. After the standard skin preparation, the abdomen of the rats was cut along the midline with a 4 cm incision. The muscular layer of the abdominal wall was drilled and a wound was created with a diameter of 6 mm using a perforator. Anti-adhesion membranes with a size of 2×2 cm were fixed at four corners around the wounds by a 6–0 surgical suture. In the control group, the abdominal cavity was closed without treatment after the wound was established. The rats were euthanized on the 7th day and the 21st day after treatment for adhesion assessment and histological analysis. A standard adhesion scoring system was used: score 0, no adhesion; score 1, mild, easily separable intestinal adhesion; score 2, moderate intestinal adhesion, separable by blunt dissection; and score 3, severe intestinal adhesion requiring sharp dissection to separate.

On the 7th day and the 21st day, the ascites of rats were extracted to detect the level of inflammatory factors. For intraperitoneal lavage, 10 mL of PBS buffer was injected into the abdominal cavity of the rats after they were anesthetized with

isoflurane. After 10 minutes, the peritoneal effusion was collected into a sterile centrifuge tube and centrifuged at 2000 rpm for 10 minutes. The supernatant was collected and stored in a cryopreservation tank at -80°C . Enzyme linked immunosorbent assay (ELISA) was used to quantitatively detect the level of inflammatory factors including TNF- α , IL-1 β , IL-6, TGF- β , IL-4 and IL-10 in the peritoneal effusion.

After the peritoneal effusion was extracted, the rats were euthanized to collect the injured tissues for pathological analysis. The tissue was fixed using 10% formaldehyde aqueous solution for 3 days, dehydrated by gradient alcohol, embedded in paraffin and cut into 6 μm sections. The slices were baked at 60°C for 1 hour on the toaster to prevent tissue shrinkage and peeling during dyeing. The sections were stained with H&E to observe the formation of adhesion and inflammatory cell infiltration. MT was used to detect the deposition of collagen during the adhesion process. All stained sections were observed using an upright microscope. Collagen volume fraction in tissues was quantified using ImageJ software.

Immunofluorescence staining was employed to detect the proportion of M2 macrophages. Briefly, the paraffin sections were dewaxed and rewatered with gradient alcohol. An EDTA antigen repair buffer (PH 9.0) was used for antigen repair in a microwave oven. After cooling, the slices were permeated by Triton X-100 penetrating liquid for 30 min and then incubated with 1% bovine serum albumin solution (BSA) for 30 min. After the sealing solution was thrown off, the first antibody against CD206 (cat. no. ab955, Abcam, 500:1) and CD68 (cat. no. 24595, CST, 500:1) were dropped on the slice and incubated in a wet box at 4°C overnight. The slices were shaken and washed for 5 min on the shaker using PBS (PH7.4) 3 times. After the sections were dried, the secondary antibodies were dropped onto the tissues, including Alexa Fluor® 488 conjugated anti-rabbit IgG (cat. no. 4412, CST, 500:1) and CY3-conjugated goat anti-mouse IgG (cat. no. S0012, Affinity, 500:1), and incubated at room temperature in the dark for 50 min. The slices were shaken and washed for 5 min on the shaker using PBS (PH7.4) 3 times. After being dried, the slices were sealed with 4',6-diamidino-2-phenylindole (DAPI, S2110, Solarbio) containing an anti-fluorescence quenching sealing agent and stored at 4°C away from light. The sections were observed using a fluorescence microscope (BX51, Olympus).

Double immunostaining of PAI-1 and t-PA was performed to investigate the expressions of PAI-1 and t-PA from adhesion-related tissues. Samples were incubated in primary antibodies overnight at 4°C , including mouse antirat PAI-1 antibody (1:100, cat. no.: 66261-1-Ig, Proteintech Group, USA) and rabbit antirat t-PA antibody (1:50, cat. no.: 10147-1-AP, Proteintech Group, USA), and then treated with Alexa Fluor 594-conjugated goat antimouse IgG (1:1000, cat. no. 8890S; Cell Signaling Technology, USA) and Alexa Fluor 488-conjugated goat antirabbit Immunoglobulin G (IgG) (1:1000, cat. no. 4412S; Cell Signaling Technology, USA). The stained samples were further incubated with DAPI for 30 min and observed using a microscope (BX53, Olympus).

Statistical analysis

All the quantitative data were recorded as mean $\pm$ standard deviation (SD). The number of samples (n) is indicated in the text or figure captions. Student's t -test was used to determine the statistical significance between two groups in GraphPad Prism 7. A p -value <0.05 was considered as statistically significant.

Conclusion

In summary, an anti-adhesion composite membrane with an immunomodulatory effect was successfully prepared in this study. Anti-inflammatory chondroitin sulfate was simply blended with PLGA to obtain the bioactive membrane through electrospinning technology. It was found that the PLGA/CS membrane could completely prevent the formation of post-operative abdominal adhesion in rats. Mechanistically, the PLGA/CS membrane significantly decreased the inflammatory response after tissue injury, accelerated the wound healing, and activated the fibrinolytic system, which together suppressed tissue adhesion. Overall, this study provided a new concept for the design of anti-adhesion polymer membranes, which added bioactivity such as anti-inflammation or pro-regeneration to the physical barrier. Given the low cost, facile large-scale manufacturing and favorable anti-adhesion function, the hybrid membrane holds great promise for clinical translation.

Author contributions

Rui Gao and Fenghui Li: conceptualization, methodology, investigation, validation, formal analysis, and writing – original draft. Yushan Zhang, Pengxu Kong and Yu Gao: methodology, investigation, validation, formal analysis and funding acquisition. Jingrong Wang, Xiang Liu and Shuangyang Li: validation, formal analysis, funding acquisition, and writing – review and editing. Liqin Jiang and Pingsheng Huang: methodology, investigation, validation, formal analysis, and writing – review and editing. Ju Zhang and Chuangnian Zhang: methodology and formal analysis. Zujian Feng: validation, formal analysis, and writing – review and editing. Weiwei Wang: conceptualization, project administration, funding acquisition and supervision.

Conflicts of interest

There are no conflicts to declare.

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